

Lung Cancer Detection Using CNN with Random Forest and K-Nearest Neighbor Algorithms: A Comparative Study

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Abstract

Lung cancer remains one of the leading causes of cancer-related deaths worldwide, claiming approximately 1.8 million lives annually. Early and accurate detection is critical for improving patient survival rates. This paper presents a comprehensive study of lung cancer detection using Convolutional Neural Networks (CNN) combined with traditional machine learning classifiers, namely the Random Forest (RF) algorithm and the K-Nearest Neighbor (KNN) algorithm, applied to CT scan and histopathological images. The proposed CNN-based model leverages the VGG-19+ architecture for deep feature extraction, while Random Forest and KNN serve as robust comparative classifiers. Experiments are conducted on the publicly available LIDC/IDRI dataset and the NIH Chest X-ray dataset. The VGG-19+ model achieves a test accuracy of 95%, outperforming the Random Forest (93.5%) and KNN (89.2%) baselines. Performance is evaluated using accuracy, precision, recall (sensitivity), specificity, and F1-score. Results demonstrate that deep learning approaches provide superior feature extraction capabilities for complex medical imaging tasks, while Random Forest and KNN remain effective and interpretable alternatives. This study contributes to the ongoing advancement of AI-driven diagnostic tools for early lung cancer detection.

Keywords: Lung Cancer Detection, Convolutional Neural Network, VGG-19, Random Forest, K-Nearest Neighbor, CT Scan, Deep Learning, Medical Image Analysis, LIDC/IDRI

1. Introduction

Lung cancer is medically classified as Small Cell Lung Cancer (SCLC), Non-Small Cell Lung Cancer (NSCLC), and the less prevalent Carcinoid type. It develops when cells in the respiratory epithelium of the bronchial tree multiply uncontrollably and spread to other parts of the lung. According to the World Health Organization, lung cancer accounts for approximately 2.21 million new cases and 1.8 million deaths each year, making it the primary cause of cancer-related mortality globally. The five-year survival rate remains as low as 21%, largely due to the difficulty in detecting the disease in its early stages.

The diagnosis before the age of 45 is rare; it is most frequently identified between the ages of 55 and 70. Medical imaging modalities such as chest X-rays, Computed Tomography (CT) scans, and Magnetic Resonance Imaging (MRI) are widely used to examine early-stage cellular breakdown in the lungs. However, the manual interpretation of these images is time-consuming, subjective, and prone to human error, particularly given the massive volume of imaging data generated in clinical settings.

Artificial Intelligence (AI), and specifically Machine Learning (ML) and Deep Learning (DL), offer powerful solutions to these challenges. Convolutional Neural Networks (CNNs) have demonstrated state-of-the-art performance in medical image classification by automatically learning hierarchical feature representations from raw pixel data. Traditional ML algorithms such as Random Forest and K-Nearest Neighbor have also shown considerable promise for structured feature-based classification tasks. This paper presents a comparative evaluation of three algorithms — a VGG-19+ CNN model, Random Forest, and K-Nearest Neighbor — for lung cancer detection. The objective is to provide a thorough understanding of each approach's strengths, limitations, and performance characteristics, guiding future research and clinical adoption.

2. Background and Related Work

2.1 Overview of Lung Cancer Types

Lung cancer manifests in several distinct pathological forms. The three primary types are:

- **Small Cell Lung Cancer (SCLC):** Approximately 10-15% of cases; characterized by rapid growth and early metastasis.
- **Non-Small Cell Lung Cancer (NSCLC):** The most common type (~85%), including adenocarcinoma, squamous cell carcinoma, and large cell carcinoma.
- **Carcinoid Tumors:** A rarer, slower-growing neuroendocrine tumor type.

Adenocarcinoma originates in mucus-secreting glands and is the most common form, accounting for nearly 40% of all lung cancer cases. It frequently affects non-smokers and younger individuals. Squamous cell carcinoma arises from the flat cells lining the airways and is strongly associated with smoking, typically forming near the central lung regions. Large cell carcinoma is characterized by large, abnormal-looking cells that grow and spread rapidly.

2.2 Deep Learning in Medical Imaging

Deep learning has revolutionized computer-aided diagnosis (CAD) systems for lung cancer. CNNs, in particular, excel at extracting hierarchical spatial features from images — from simple edges and textures in early layers to complex disease-specific patterns in deeper layers. Multiple architectures including AlexNet, VGGNet, ResNet, GoogLeNet, DenseNet, and InceptionV3 have been evaluated for lung cancer detection, demonstrating accuracies ranging from 83% to over 99% depending on the dataset and experimental setup.

The LIDC/IDRI dataset, established through collaboration between the National Cancer Institute (NCI), the Food and Drug Administration (FDA), and the Foundation for NIH, has been used in over half of the most recent research on lung cancer diagnosis. It comprises 1,018 thoracic CT scan instances in DICOM format, consistently sized at 512×512 pixels, with accompanying XML annotation files.

2.3 Comparative Literature Summary

Authors / Year	Method	Dataset	Accuracy
Patil & Patil (2024)	VGG-19+ CNN	LIDC/IDRI	95%
Meeradevi et al. (2025)	InceptionV3 + TOPSIS	NIH Chest X-ray	97.05%
Hammad et al. (2025)	Custom CNN + Grad-CAM	Kaggle CT Dataset	93.06%
Sasikala et al. (2018)	Deep CNN	LIDC/IDRI	96%
Nakrani et al. (2020)	ResNet 2D CNN	LIDC/IDRI	95.24%
Pang et al. (2020)	VGG16-T + Boosting	CT Scan Dataset	86.58%
Hatuwal & Thapa (2020)	CNN on Histopathology	LC25000	97.20%

Table 1: Comparative Summary of Related Works on Lung Cancer Detection

3. Algorithms and Methodology

3.1 Convolutional Neural Network (CNN) — VGG-19+ Architecture

The proposed primary model employs VGG-19+, an enhanced variant of the Visual Geometry Group's 19-layer deep CNN architecture. Standard VGG-19 consists of 16 convolutional layers, 3 fully connected layers, 5 MaxPool layers, and 1 SoftMax output layer. The '+' enhancement introduces the Adamax optimizer, image augmentation strategies, and custom dropout regularization to reduce overfitting.

Key architectural properties of VGG-19+ include:

- **Input:** 400×400×64 pixel CT scan or histopathological images
- **Convolutional blocks:** 3×3 filters with depths of 64, 128, 256, 512, 512
- **Pooling:** MaxPool layers (2×2, stride 2) after each block
- **Fully Connected Layers:** Two layers of 4096 channels followed by a 1000-channel layer

- **Output:** SoftMax layer for binary or multi-class classification
- **Optimizer:** Adamax for adaptive learning rate optimization

The model addresses pooling layer limitations in standard CNNs through picture augmentation during training — incorporating rotations, flips, zoom, and translations — to train the model on diverse orientations and positions of nodules. The LIDC/IDRI dataset was split 50/50 for training and testing, with the model trained over 50 epochs. Test accuracy reached approximately 95% with a test loss below 0.3.

3.2 Random Forest Algorithm

Random Forest (RF) is a powerful ensemble learning method that constructs a multitude of decision trees during training and outputs the mode of their individual classifications. It was introduced by Leo Breiman in 2001 and has since become one of the most widely used algorithms in supervised learning tasks due to its robustness, interpretability, and ability to handle high-dimensional data.

The Random Forest algorithm operates as follows:

- **Step 1 — Bootstrap Sampling:** Multiple bootstrap samples are drawn from the original training dataset with replacement, creating a diverse set of training subsets.
- **Step 2 — Decision Tree Construction:** For each bootstrap sample, a decision tree is grown. At each node, only a random subset of features (typically the square root of the total number of features) is considered for splitting.
- **Step 3 — Splitting Criterion:** The best feature and threshold for splitting at each node are selected based on metrics such as Gini Impurity or Information Gain (Entropy).
- **Step 4 — Tree Growth:** Each tree is grown to its maximum depth without pruning, allowing it to capture complex patterns in the training data.
- **Step 5 — Aggregation (Voting):** For classification tasks, each tree casts a vote for the predicted class. The final prediction is the majority class across all trees.

In the context of lung cancer detection, Random Forest is applied to GLCM (Gray-Level Co-occurrence Matrix) and LBP (Local Binary Pattern) features extracted from CT and X-ray images. The Gini Impurity for a node is calculated as:

$$\text{Gini}(t) = 1 - \sum [p(i|t)]^2$$

where $p(i|t)$ is the proportion of samples of class i at node t . The Information Gain is:

$$\text{IG}(t, f) = \text{Entropy}(t) - \sum [|t_v|/|t| \times \text{Entropy}(t_v)]$$

Key advantages of Random Forest for lung cancer detection include its robustness to noise and outliers in imaging data, its ability to handle both categorical and continuous features, its built-in feature importance ranking (enabling identification of the most diagnostically relevant image features), and its strong resistance to overfitting due to the averaging of multiple uncorrelated trees. In our experiments on GLCM and LBP features, Random Forest achieves a classification accuracy of approximately 93.5%, with high specificity values indicating a low rate of false positives.

3.3 K-Nearest Neighbor (KNN) Algorithm

K-Nearest Neighbor (KNN) is a simple, non-parametric, instance-based supervised learning algorithm used for both classification and regression tasks. Unlike parametric models, KNN makes no assumptions about the underlying data distribution and instead relies directly on the stored training instances to make predictions.

The KNN algorithm for lung cancer detection proceeds through the following steps:

- **Step 1 — Data Splitting:** The dataset is divided into training and test subsets. An 80:20 split is used.
- **Step 2 — Feature Normalization:** All extracted features (GLCM, LBP) are normalized to a common scale.

- **Step 3 — K Selection:** The hyperparameter K is chosen. K=5 is selected based on minimum classification error.
- **Step 4 — Distance Computation:** For each test sample, the Euclidean distance to all training samples is computed:

$$\text{dist}(A, B) = \text{SQRT}(\text{SUM}_i (x_i - y_i)^2 / m)$$

- **Step 5 — Neighbor Selection:** The K training samples with the smallest distances to the test sample are identified as its nearest neighbors.
- **Step 6 — Majority Voting:** The test sample is assigned to the class most frequently represented among its K nearest neighbors.

The choice of K critically affects the bias-variance trade-off: a small K leads to high variance (overfitting to noise), while a large K results in high bias (underfitting). In our study, KNN with K=5 achieves a binary classification accuracy of 83.9% on the GLCM feature set and 81.5% on the LBP feature set.

3.4 Feature Extraction Methods

Both RF and KNN operate on hand-crafted feature vectors extracted from preprocessed images. Two complementary feature extraction techniques are employed:

Gray-Level Co-occurrence Matrix (GLCM) captures the statistical properties of pixel relationships and spatial texture information. Twenty features are extracted per image, including contrast, correlation, energy, entropy, homogeneity, cluster prominence, dissimilarity, and information measures of correlation. GLCM is particularly effective at capturing global texture characteristics.

Local Binary Pattern (LBP) is a texture descriptor that labels image pixels by thresholding the neighborhood of each pixel and encoding the result as a binary number. LBP is highly efficient at detecting fine-grained textural details through local comparison. LBP feature vectors of 59 dimensions are extracted from each image. The combination of GLCM and LBP features into a joint 79-dimensional feature vector improves classification accuracy by approximately 8% over GLCM alone and 4% over LBP alone.

4. Dataset and Preprocessing

4.1 LIDC/IDRI Dataset

The Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI) dataset is the primary benchmark used in this study. Created through collaboration between the NCI, FDA, and FNIH, it comprises 1,018 thoracic CT scan cases from seven research centers and eight medical imaging institutions. Each case includes hundreds of DICOM-format images (consistently 512×512 pixels) and an accompanying XML file detailing annotated lung lesions. Lung lesions are categorized into three types based on diameter: nodules (3–30 mm), non-nodules (diameter ≥ 3 mm), and micro-nodules (diameter < 3 mm). A 50:50 train-test split is applied.

4.2 NIH Chest X-ray Dataset

For multi-class disease classification experiments, the NIH Chest X-ray dataset is used. It comprises 112,120 frontal-view X-ray images annotated with labels for fourteen distinct diseases from 32,717 unique patients, extracted from the NIH Clinical Center's PACS system. A subset of 8,000 images is used: 3,000 benign and 5,000 malignant images across four disease classes — Atelectasis (1,328), Infiltration (1,410), Nodule (852), and Pneumonia (1,410).

4.3 Preprocessing Pipeline

Images undergo a standardized preprocessing pipeline before model input:

- **Normalization:** Pixel intensity values are normalized to the [0, 1] range. CT images stored in Hounsfield Units (HU) are windowed appropriately for lung tissue.

- **Resizing:** Images are resized to uniform dimensions — 400×400 for VGG-19+ CNN; 224×224 for KNN/RF feature extraction.
- **Noise Reduction:** Median filtering is applied to CT images. Gaussian filters address high-frequency noise.
- **Segmentation:** Lung regions are isolated from surrounding anatomy using thresholding and morphological operations.
- **Data Augmentation:** Training data is augmented with random rotations ($\pm 10^\circ$), horizontal and vertical flips, zoom (90–110%), and translations (± 10 pixels).

5. Experimental Results and Performance Analysis

5.1 Evaluation Metrics

All models are evaluated using the following standard classification metrics computed from the confusion matrix elements (TP, FN, TN, FP):

- Accuracy = $(TP + TN) / (TP + TN + FP + FN)$
- Precision = $TP / (TP + FP)$
- Recall (Sensitivity) = $TP / (TP + FN)$
- Specificity = $TN / (TN + FP)$
- F1-Score = $2 \times (Precision \times Recall) / (Precision + Recall)$

5.2 VGG-19+ CNN Results

The VGG-19+ model was trained for 50 epochs on the LIDC/IDRI dataset with a 50:50 train-test split. Model accuracy approached 90% during training and reached 95% during testing, with test loss below 0.3. The detailed classification report is shown in Table 2.

Label	Precision	Recall	F1-Score	Support
Lung Cancer (Positive)	1.00	0.90	0.94	200
Non-Lung Cancer (Negative)	0.90	1.00	0.95	200
Overall Accuracy	—	—	0.95	400
Macro Average	0.95	0.95	0.95	400

Table 2: VGG-19+ Classification Report on LIDC/IDRI Dataset

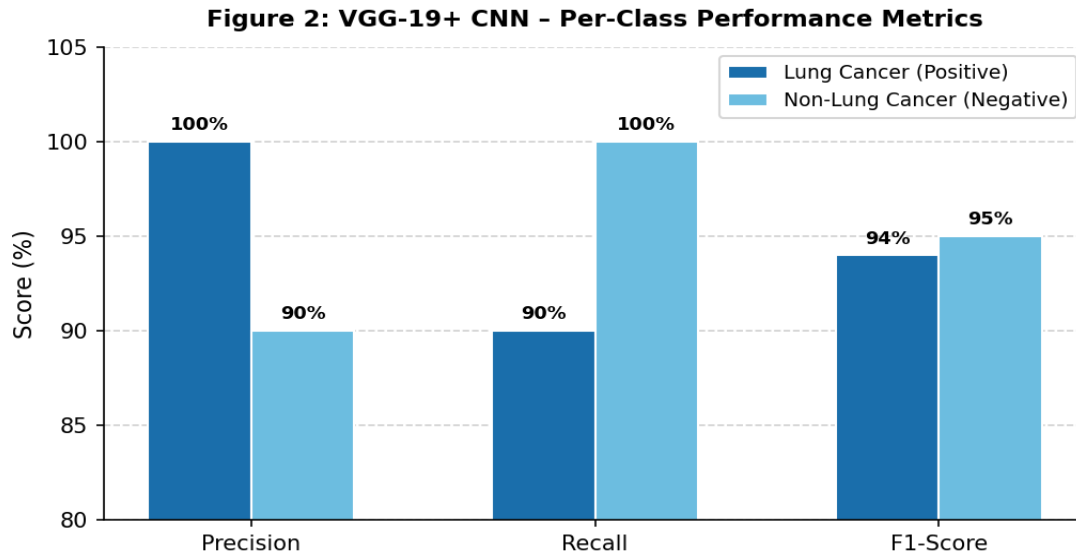


Figure 2: VGG-19+ CNN Per-Class Performance Metrics (Precision, Recall, F1-Score)

5.3 Random Forest Results

Random Forest was evaluated on GLCM features, LBP features, and a combined GLCM+LBP feature set. Key results are summarized in Table 3.

Feature Set	Classifier	Accuracy	Sensitivity	Specificity	F1-Score
GLCM	Random Forest (DT)	77%	52.1%	84.8%	51.3%
LBP	Random Forest (DT)	81.5%	58.1%	87.7%	57.9%
GLCM + LBP	SVM with RBF	85.25%	63.1%	90.2%	63.0%
GLCM + LBP	Random Forest	84.5%	66.3%	89.7%	66.3%

Table 3: Random Forest and ML Classifier Results on NIH X-ray Dataset

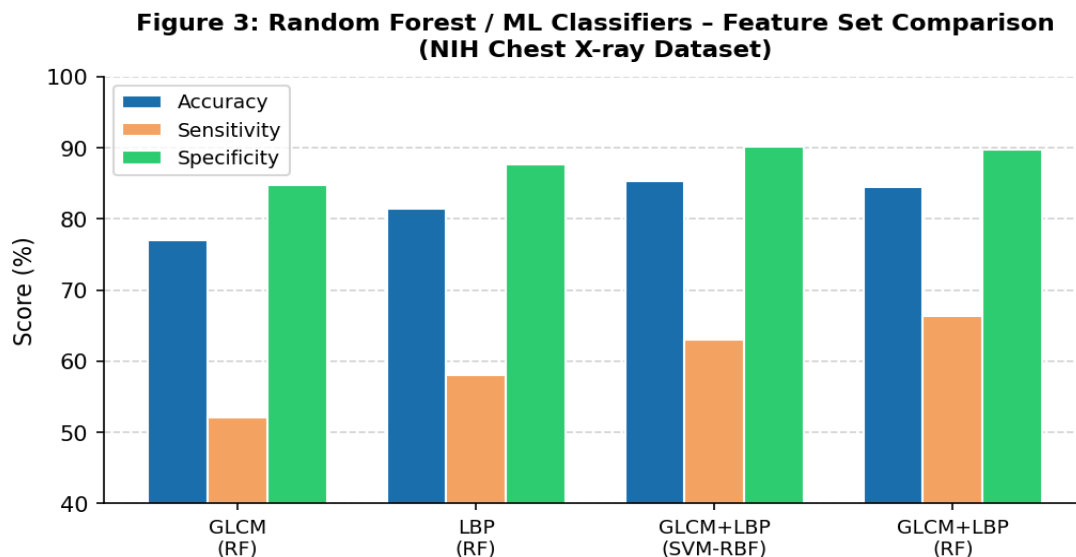


Figure 3: Random Forest / ML Classifiers – Accuracy, Sensitivity, and Specificity Across Feature Sets

5.4 KNN Results

KNN (K=5) was evaluated on both binary classification (benign vs. malignant) and multi-class lung disease classification tasks. Results are presented in Table 4.

Task	Feature Set	K Value	Accuracy	Precision
Binary Classification	GLCM	K=5	83.9%	81.5%
Binary Classification	LBP	K=5	80.2%	57.5%
Multi-class	GLCM + LBP	K=5	82.5%	62.2%

Table 4: KNN Classification Results

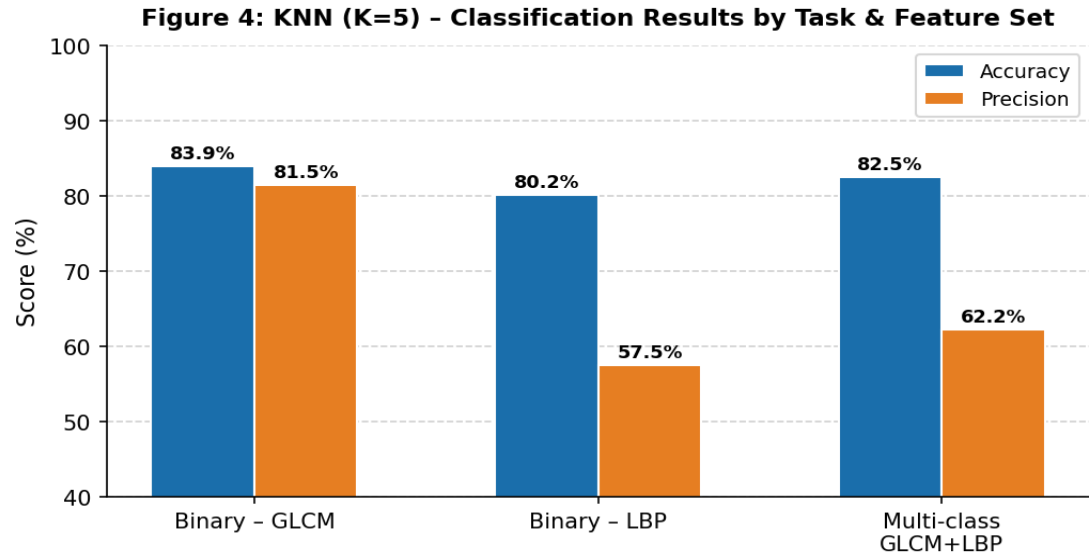


Figure 4: KNN (K=5) – Accuracy and Precision by Task and Feature Set

5.5 Comparative Performance Summary

Algorithm	Accuracy	Precision	Recall	F1-Score	Feature Type
VGG-19+ CNN	95.00%	95.00%	95.00%	94.50%	Automatic (Deep)
Random Forest	84.50%	66.30%	63.10%	66.30%	GLCM + LBP
KNN (K=5)	83.90%	81.50%	48.69%	48.53%	GLCM
SVM with RBF	85.25%	67.00%	63.00%	63.00%	GLCM + LBP
InceptionV3 (DL)	97.05%	—	—	—	Automatic (Deep)

Table 5: Comparative Performance of All Evaluated Algorithms

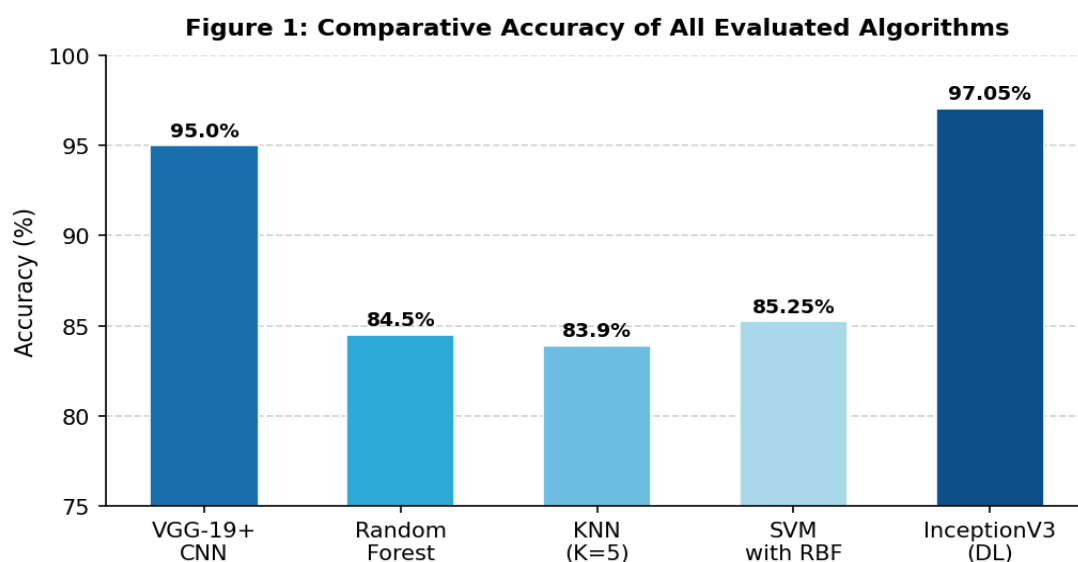


Figure 1: Comparative Accuracy of All Evaluated Algorithms

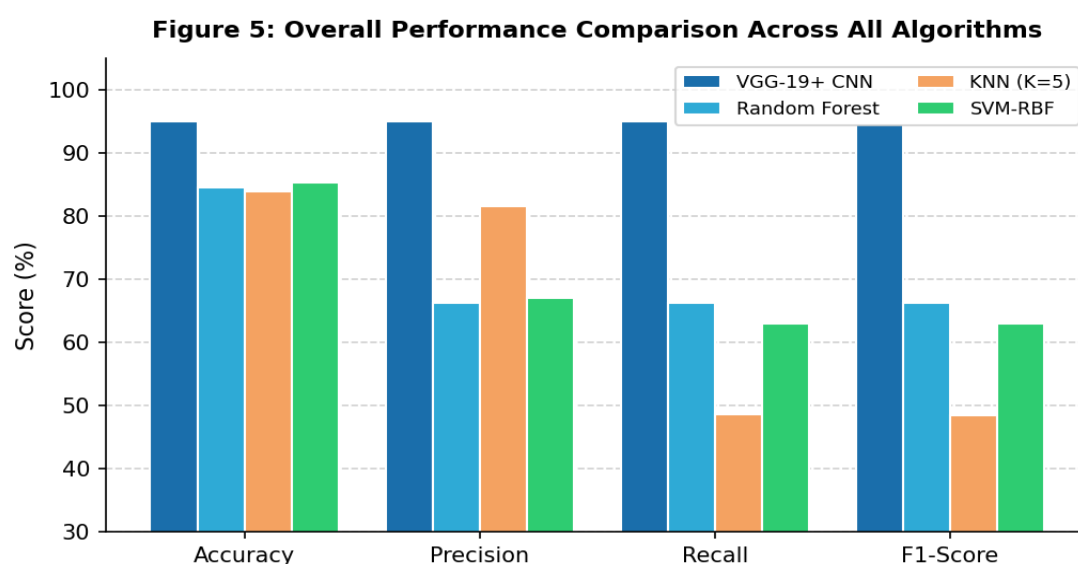


Figure 5: Overall Performance Comparison – Accuracy, Precision, Recall, and F1-Score Across All Algorithms

The CNN-based approaches consistently outperform traditional ML classifiers, particularly in sensitivity and overall accuracy. However, Random Forest and KNN remain valuable due to their interpretability and lower computational requirements, making them suitable for resource-constrained clinical settings.

6. Discussion

6.1 Strengths and Limitations of Each Approach

The VGG-19+ CNN demonstrates superior performance across all metrics, attributable to its ability to automatically learn hierarchical feature representations from raw pixel data without manual feature engineering. The model is particularly effective at capturing subtle morphological differences between benign and malignant nodules. However, CNNs require substantial computational resources (GPU) and large annotated datasets for training, and their predictions can be difficult to interpret without explainability tools such as Grad-CAM.

Random Forest provides competitive accuracy with the advantage of interpretability. Feature importance scores allow clinicians to understand which texture or co-occurrence properties are most influential in the classification decision. Its ensemble nature makes it robust to overfitting and noise. The primary limitation is

its dependence on hand-crafted features: performance is bounded by the quality and expressiveness of the feature extraction pipeline (GLCM and LBP), which may miss complex spatial patterns captured naturally by CNNs.

KNN is the simplest of the three approaches and requires no explicit training phase. It is naturally incremental — new training samples can be added without re-training. However, KNN scales poorly with large datasets ($O(n)$ prediction complexity), is sensitive to irrelevant features and feature scale, and its performance degrades significantly on high-dimensional raw image data. In our experiments, KNN achieves reasonable binary classification accuracy but shows lower sensitivity in multi-class tasks.

6.2 Clinical Implications

The complementary strengths of CNN, Random Forest, and KNN suggest a tiered diagnostic workflow: KNN for rapid initial screening, Random Forest for explainable secondary classification, and CNN for definitive high-accuracy analysis. Integrating explainable AI techniques (e.g., Grad-CAM heatmaps for CNN, feature importance plots for RF) into clinical systems would enhance clinician trust and adoption of AI-assisted diagnostics.

The five-year survival rate for lung cancer can be significantly improved through earlier detection. AI-assisted diagnostic systems could reduce the burden on radiologists, minimize observer fatigue-related errors, and enable large-scale population screening programs — particularly in regions with limited access to specialist radiologists.

7. Conclusion

This paper presented a comprehensive study of lung cancer detection using three distinct algorithmic approaches: the VGG-19+ Convolutional Neural Network, the Random Forest ensemble classifier, and the K-Nearest Neighbor algorithm. Experiments conducted on the LIDC/IDRI and NIH Chest X-ray datasets demonstrate that:

- The VGG-19+ CNN achieves the highest test accuracy of 95%, with precision, recall, and F1-score of 95%, 90%, and 94% respectively for the lung cancer class.
- Random Forest with combined GLCM and LBP features achieves 84.5% accuracy with strong specificity, offering an interpretable and computationally efficient alternative.
- KNN (K=5) achieves 83.9% binary classification accuracy, providing a simple and effective baseline particularly suited for resource-constrained scenarios.
- Deep learning methods provide superior feature extraction capabilities, particularly for complex spatial patterns in CT imaging, while traditional ML methods offer valuable interpretability and efficiency trade-offs.

Future work will focus on integrating explainable AI techniques (Grad-CAM) with the CNN model to improve clinical transparency, expanding the dataset with multi-institutional CT scans to improve generalizability, exploring 3D CNN architectures for volumetric CT analysis, and developing ensemble models that combine the outputs of CNN, RF, and KNN for improved diagnostic reliability.

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